46-926, Fall 2020: Homework #2

Due 3:20pm, November 11, 2019.

Homework should be submitted electronically on canvas before the deadline. Please submit your homework as a single PDF file.

Do all computational problems in python unless otherwise noted. Please post any questions on the Piazza discussion board.

1. This is just a simple problem to better explore some of the ideas of variance, eigenvalues/vectors, and multivariate distributions. We are going to keep coming back to these ideas, and will need an intuition. In this problem, we use simulation to explore these ideas for the multivariate normal distribution.

   Let μ be a length p vector, and Σ be a $p \times p$ matrix.

   You can generate a random multivariate normal vector $X \sim N(\mu, \Sigma)$ using `numpy.random.multivariate_normal` or `scipy.stats.multivariate_normal`.

   (a) Define matricies `mu` and `Sigma` to be

   $$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \qquad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

   where $\rho = 0.6$. You can use the code:

   (a) **Draw** a sample of size $n = 100$ from a multivariate normal with mean μ and covariance Σ. **Confirm** that the output is a 100×2 matrix. I will refer to this as `X` below. Each row represents a sample from your 2-dimensional multivariate normal distribution.

   Plot the first column of `X` against the second column of `X`. Set the x-axis limits to $[-2, 4]$ and the y-axis limits to $[-1, 5]$. Add a large red dot in the middle to represent your mean μ.

   Remake the same plot but using $\rho = 0.9$ in Σ. What changes?

   (b) Now we will look at the marginal distributions of your data. **Plot** a histogram of the first column of your data (set `density=True` when you call `hist`, so it represents a density works for the next step). **Overlay** a normal curve on the same plot with mean `mu[1]` and variance `Sigma[1,1]`.

   You should see that this marginal distribution looks quite Gaussian!

(c) Let $v = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, the diagonal vector of length 1.

Multiplying a point x_i by v, $x_i^T v$, gives the length of the projection of x_i onto the vector v. You can compute Xv (where this is matrix multiplication) to give the projection of every data point onto `v`. **Plot** a histogram of Xv.

This should again look Gaussian. It turns out that all projections of a multivariate Gaussian are Gaussian! How does the variance of this gaussian compare to the one from the previous step?

(d) For both settings in part (a), Compute the eigenvectors of Σ. Call these `v1` and `v2` respectively.

Starting with the plots from (a), **draw** `v1` and `v2` as lines starting at the mean (do this for both plots). Make each line as long as $2\sqrt{\lambda_i}$ where λ_i is the corresponding eigenvalue for v_i. What do you notice?

(g) Form a matrix Z that has columns Xv_1 and Xv_2, the projections of your samples onto each of the eigenvectors.

Plot the first column of Z against the second column of Z. What do you notice?

Compute the correlation of the first column of X with the second column of X. Similarly, compute the correlation of the first column of Z with the second column of Z. What do you notice?

2. Inference after selection. Next class, we'll be starting to encounter methods that carry out *variable selection* in regression, meaning that the estimated model only uses a subset of the available variables. However, you've already seen some approaches with this property when you discussed stepwise regression in previous classes. One temptation that people experience when using methods of this nature is to also carry out statistical inference on the coefficients that are selected.

   In this problem, we are going to simulate a very simple version of this problem, and see what happens. As in some other settings, we are going to abstract away the "regression" aspect, and simply consider normally distributed quantities. Like we will discuss in class, this system will turn out to exhibit the same behavior as regression.

   Suppose that you are looking at 100 different quantities $Y_1, \ldots, Y_{100}$ (you can imagine excess returns of different assets, if you like). For simplicity, we will pretend that each quantity is normally distributed with its own mean and variance 1, $Y_i \sim N(\mu_i, 1)$. For your purpose, you'd like to pick an index that has good performance and get a confidence interval for its mean.

Classic statistics tells us that in general:

- Our MLE for μ_i is Y_i.
- A 95% confidence interval for μ_i should be given by $Y_i \pm 1.96$.

(a) First, let's make sure that our usual understanding of statistics works. Simulate the above setup 100 times, setting all $\mu_i = 0$. Suppose that we knew we were interested in Y_1 ahead of time. For each simulation, calculate the MLE and confidence interval for μ_1.

Plot the distribution of $\widehat{\mu_1}$ estimates across simulations. Calculate the fraction of confidence intervals that cover $\mu_1 = 0$.

(b) Now suppose instead that we choose to look at the index with the biggest Y_i, so that we pick a good performing asset. Repeat your 100 simulations with all $\mu_i = 0$. This time, pick the i such that Y_i is largest, and compute the MLE for μ_i and a confidence interval for μ_i.

Plot the distribution of $\widehat{\mu_i}$ estimates across simulations. Calculate the fraction of confidence intervals that cover their respective $\mu_i = 0$.

This is an illustration with a very simple toy example, but the same thing will happen in general when you select your hypothesis or model based on your data and then ask standard inferential questions about that hypothesis/model.

3. Given a response vector $y \in \mathbb{R}^n$, predictor matrix $X \in \mathbb{R}^{n\times p}$, and tuning parameter $\lambda \geq 0$, recall the ridge regression estimate

$$\widehat{\beta}^{\text{ridge}} = \underset{\beta\in\mathbb{R}^p}{\operatorname{argmin}} \; \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2.$$

(a) Show that $\widehat{\beta}^{\text{ridge}}$ is simply the vector of linear regression coefficients from regressing the response $\tilde{y} = \begin{bmatrix} y \\ 0 \end{bmatrix} \in \mathbb{R}^{n+p}$ onto the predictor matrix $\tilde{X} = \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \in \mathbb{R}^{(n+p)\times p}$, where here $0 \in \mathbb{R}^p$, and $I \in \mathbb{R}^{p\times p}$ is the identity matrix.

(b) Show that when $\lambda > 0$, the matrix $\tilde{X}$ always has full column-rank, i.e., its columns are always linearly independent, regardless of the columns of X. This means that the ridge regression estimate is always unique, for any matrix of predictors X!

(c) Write out an explicit formula for $\widehat{\beta}^{\text{ridge}}$ involving X, y, λ. Conclude that for any $a \in \mathbb{R}^p$, the estimate $a^T \widehat{\beta}^{\text{ridge}}$ is a linear function of y. **Hint:** Take advantage of part (a), instead of doing any calculus or real work.

Sidenote: Among other things, it is interesting that the ridge regression solution is a linear function of Y, because this means that if it were unbiased, it would be covered by the Gauss-Markov theorem. This means that if it were not biased, it could not have lower variance than Least squares, so would lose in prediction error.

It is also worth noting that unlike Ridge regression, the Lasso (next class) is not a linear function of Y.

4. **Fun with the SVD!** The *Singular Value Decomposition* (SVD) is very useful for understanding many objects in machine learning. In this problem, you will see that it gives insight into ridge regression (and linear regression). We will see it appear again throughout the semester.

Let $X \in \mathbb{R}^{n \times p}$ be our data matrix. For this problem, we assume that $n > p$ and that X is full column rank (though this is not necessary for SVD). We can write

$$X = UDV^T$$

where $U \in \mathbb{R}^{n \times p}$ and $V \in \mathbb{R}^{p \times p}$ are orthogonal and $D \in \mathbb{R}^{p \times p}$ is diagonal. (You may have seen the SVD written with different dimensions in the past. Writing it like this, with U being non-square instead of D, is sometimes called the "thin" version of the SVD. Everything for this problem will work out a little prettier with this version.) Let $d_1, \ldots, d_p$ be the p diagonal elements of D.

Note: There are many hints on linear algebra at the end of this problem.

(a) Consider least squares linear regression. We know that $\widehat{Y} = X\widehat{\beta} = X(X^TX)^{-1}X^TY$. Replace X with its SVD, and simplify as much as possible. How can you interpret this in terms of the columns of U and/or V? (see hints)
(Fun ungraded question to think about: if you already knew the SVD of X, what is the computation time to run a regression?)

(b) Now consider ridge regression with fixed λ. We showed in the previous problem that

$$\widehat{Y}_{ridge} = X\widehat{\beta}_{ridge} = X(X^TX + \lambda I)^{-1}X^TY.$$

Again, replace X with its SVD and simplify as much as possible. You will wind up with a matrix product that looks more complicated than in (a).

Hints:

- You will find it very useful to to replace I with VV^T at some point. This will let you factor and invert, even though it doesn't feel like a simplification.
- You will know when you are done simplifying because your final answer will have Us and Ds, but no Vs.

(c) Try rewriting the product from (b) as a sum. You should be able to obtain a sum of the form:

$$\widehat{Y}_{ridge} = \sum_{j=1}^{p} (\textbf{A scalar you must find}) \cdot u_j u_j^T y$$

where u_j is the j^{th} column of U. Compare this to (b):

- If $\lambda = 0$, do you get the same answer as linear regression (part b)?
- If $\lambda > 0$, what is the effect of the scalar you had to find? Which terms does lambda shrink the most, those with large d_j or those with small d_j? How would you interpret that in terms of the amount of variance in each direction? (See PCA hint below.)

Hints for Problem 4:

- By orthogonality, $U^TU = V^TV = I_p$, the $p \times p$ identity matrix. Also, $VV^T = I_p$.
- D is a diagonal matrix, so $D^T = D$ and $DD = D^2$ is just a diagonal matrix with $d_1^2, \ldots, d_p^2$ on the diagonal.
- UU^T is *not* the identity. What is it? (Hints: it looks a lot like quantities that appeared in our PCA slides, think in terms of the columns of U).
- It turns out that the columns of V are the PCA directions for X, that $\frac{1}{n}d_j^2$ is the variance in the v_j direction, and that the i^{th} entry of u_j (column) is the (normalized) length of the projection of x_i onto v_j. We will see that important regression quantities can be expressed in terms of these quantities.
- If any square matrices $A, B \in \mathbb{R}^{p \times p}$ are invertable, then $(AB)^{-1} = B^{-1}A^{-1}$. In particular, for V above and D an invertable matrix, $(VDV^T)^{-1} = VD^{-1}V^T$ (since V^T is the inverse of V, and vice versa).
- If D were a diagonal matrix, then $UDU^TY = \sum_{i=1}^{n} D_{ii}u_iu_i^T y$.